1. This question carries 40% of the mark.

(a) Consider each of the following continuous-time signals. For each case, specify if the signal is causal/non-causal, periodic/non-periodic, odd/even. If the signal is periodic specify its period.

(i) $x(t) = |\sin(2\pi t)|$ **[2]**

Answer

Non-causal, periodic with period $T = 1/2$, even

(ii) $x(t) = e^{t^2}$ **[2]**

Answer

Non-causal, non-periodic, even

(b) Consider the signal

$$x(t) = \begin{cases} 1 - 2t, & 0 \le t \le 1 \\ 0, & \text{otherwise} \end{cases}$$

Now sketch each of the following signals and describe briefly in words how each of the signals can be derived from the original signal $x(t)$.

(i) $x\left(-\frac{t}{5} + 1\right)$ **[2]**

(ii) $x(3t + 5)$ **[2]**

Answer for (i) and (ii)

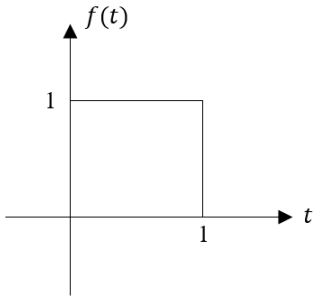
$$x\left(-\frac{t}{5} + 1\right)$$

$$= \begin{cases} 1 - 2\left(-\frac{t}{5} + 1\right) = \frac{2t}{5} - 1, & 0 \le -\frac{t}{5} + 1 \le 1 \Rightarrow -5 \le -t \le 0 \Rightarrow 0 \le t \le 5 \\ 0 & \text{otherwise} \end{cases}$$

$$x(3t + 5)$$

$$= \begin{cases} 1 - (3t + 5) = -3t - 4, & 0 \le 3t + 5 \le 1 \Rightarrow -5 \le 3t \le -4 \Rightarrow -\frac{5}{3} \le t \le -\frac{4}{3} \\ 0 & \text{otherwise} \end{cases}$$

(c) By applying the property of convolution, find the system's output (i.e., zero-state response) given that the impulse response is $h(t) = e^{-2t}u(t)$ and that the input $f(t)$ is as shown in Figure 1(c) below.



[6]

Figure 1(c)

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Answer

$h(t) = e^{-2t}u(t)$

$f(t) = \begin{cases} 1, & 0 \le t \le 1 \\ 0 & \text{otherwise} \end{cases}$

$y(t) = h(t) * f(t)$

$= \int_{-\infty}^{\infty} h(\tau)f(t-\tau)d\tau \Rightarrow y(t) = \int_{-\infty}^{\infty} e^{-2\tau}u(\tau)f(t-\tau)d\tau = \int_{0}^{\infty} e^{-2\tau}f(t-\tau)d\tau$

$f(t-\tau) = \begin{cases} 1, & 0 \le t-\tau \le 1 \Rightarrow -1 \le \tau - t \le 0 \Rightarrow t-1 \le \tau \le t \\ 0 & \text{otherwise} \end{cases}$

For $t < 0$ the convolution is zero.

For $t - 1 \le 0 \le t$ which means $0 \le t \le 1$ we have

$y(t) = \int_0^t e^{-2\tau}d\tau = \frac{1}{2} - \frac{1}{2}e^{-2t}$

For $t \ge 1$ we have $y(t) = \int_{t-1}^{t} e^{-2\tau}d\tau = \frac{1}{2}e^{-2(t-1)} - \frac{1}{2}e^{-2t} = \frac{1}{2}e^{-2t}(e^2 - 1)$

Therefore, the convolution is

$$y(t) = \begin{cases} \frac{1}{2} - \frac{1}{2}e^{-2t}, & 0 \le t \le 1 \\ \frac{1}{2}e^{-2t}(e^2 - 1) & t \ge 1 \end{cases}$$

(d) (i) Use the time-shifting property to show that if $x(t) \Leftrightarrow X(\omega)$, then $x(t+T) + x(t-T) \Leftrightarrow 2X(\omega)\cos T\omega$. **[3]**

Answer

$\mathcal{F}\{x(t+T)\} = X(\omega)e^{j\omega T}$

$\mathcal{F}\{x(t-T)\} = X(\omega)e^{-j\omega T}$

Hence,

$x(t+T) + x(t-T) \Leftrightarrow 2X(\omega)\cos T\omega$

(ii) Use the result in (d) (i) above to find the Fourier transform of the signal shown in **Figure (d)(ii)** below.

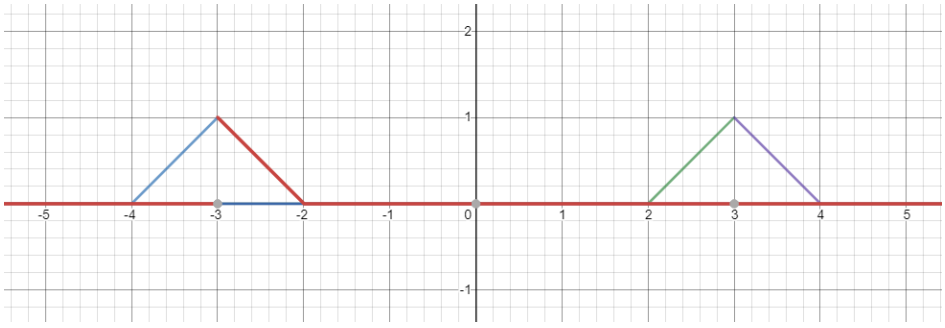


Figure 1(d)(ii)

Use the fact that $\Delta\left(\frac{t}{\tau}\right) \Leftrightarrow \frac{\tau}{2}\text{sinc}^2\left(\frac{\omega\tau}{4}\right)$ with

$$\Delta(x) = (1 - 2|x|)(u\left(x + \frac{1}{2}\right) - u\left(x - \frac{1}{2}\right))$$

[3]

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Answer

In the particular case, $\tau = 2$ and $x(t) = \Delta\left(\frac{t}{2}\right) \Leftrightarrow \text{sinc}^2\left(\frac{\omega}{2}\right)$.

Furthermore, $T = 3$.

Therefore,

$x(t+3) + x(t-3) \Leftrightarrow 2X(\omega)\cos 3\omega = 2\text{sinc}^2\left(\frac{\omega}{2}\right)\cos 3\omega$

(e) The output $y(t)$ of a continuous-time LTI system is related to the input $x(t)$ by the differential equation

$$\frac{dy(t)}{dt} + 4y(t) = 2x(t)$$

Sketch the Bode plots of the system. Detailed explanation is required. **[5]**

Answer

The transfer function is:

$$H(s) = \frac{2}{s+4}$$

We substitute $s = j\omega$. Then, we obtain $H(j\omega) = \frac{2}{j\omega+4} = \frac{1/2}{\frac{j\omega}{4}+1}$.

Amplitude response: $|H(j\omega)| = \frac{1/2}{\sqrt{\frac{\omega^2}{16}+1}}$.

Phase response: $\angle H(j\omega) = \Phi(\omega) = -\tan^{-1}\left(\frac{\omega}{4}\right)$

The log amplitude is $20\log\left(\frac{1}{2}\right) - 20\log\left|1 + \frac{j\omega}{4}\right| = -20\log 2 - 20\log\left|1 + \frac{j\omega}{4}\right|$

- $\omega \ll 4 \Rightarrow -20\log 2 - 20\log\left|1 + \frac{j\omega}{4}\right| \approx -20\log 2 - 20\log 1 = -20\log 2 = -6.02$
- $\omega \gg 4 \Rightarrow -20\log 2 - 20\log\left|1 + \frac{j\omega}{4}\right| \approx -20\log 2 - 20\log\left(\frac{\omega}{4}\right) = -20\log\omega + 20\log 2$

This represents a straight line (when plotted as a function of u, the log of ω) with a slope of $-20dB/$decade or $-20(0.301) = -6.02dB/$octave.

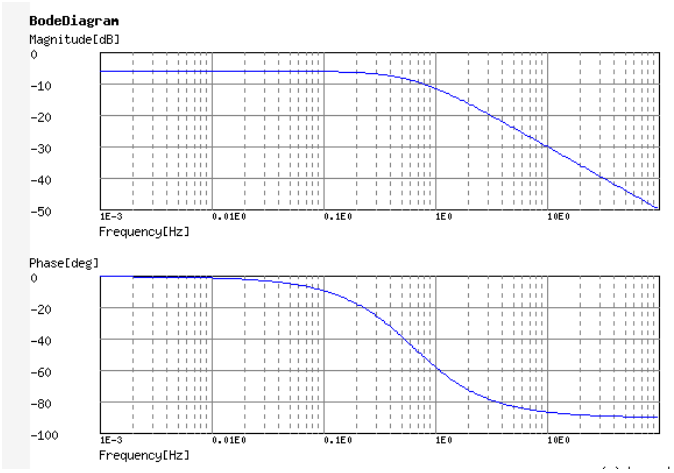
The exact log amplitude for this pole is:

$$-20\log\left|1 + \frac{j\omega}{2}\right| = -20\log\left(1 + \frac{\omega^2}{4}\right)^{\frac{1}{2}} = -10\log\left(1 + \frac{\omega^2}{4}\right)$$

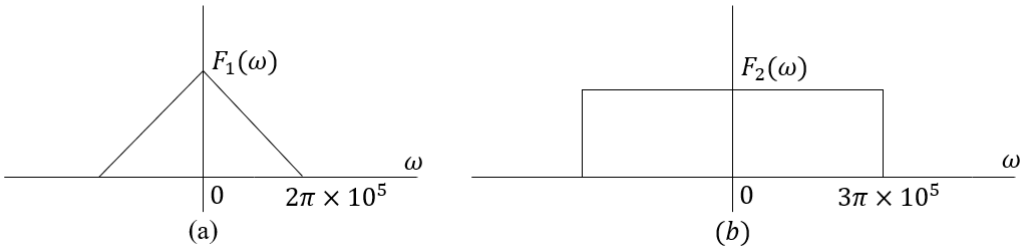
For the phase we have:

- $\angle H(j\omega) = -\angle\left(1 + \frac{j\omega}{4}\right) = -\tan^{-1}\left(\frac{\omega}{4}\right)$.
- $\omega \ll 4 \Rightarrow \frac{\omega}{4} \ll 1 \Rightarrow -\tan^{-1}\left(\frac{\omega}{4}\right) \approx 0$
- $\omega \gg 4 \Rightarrow \frac{\omega}{4} \gg 1 \Rightarrow -\tan^{-1}\left(\frac{\omega}{4}\right) \approx -90^{\text{o}}$

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(f) Figures (a) and (b) below show the Fourier spectra of signals $f_1(t)$ and $f_2(t)$. Determine the Nyquist sampling rates for signals $f_1(t)$ and $f_2(t)$. (Hint: Use the frequency convolution and the width property of the convolution.)



(i) $f_1(t)$ **[2]**

Answer

Bandwidth of $f_1(t)$ is 100kHz and Nyquist rate is 200kHz.

(ii) $f_2(t)$ **[2]**

Answer

Bandwidth of $f_2(t)$ is 150kHz and Nyquist rate is 300kHz.

(iii) $f_2^2(t)$ **[2]**

Answer

The Fourier Transform of $f_2^2(t)$ is $\frac{1}{2\pi}F_2(\omega) * F_2(\omega)$. From the property regarding the size of convolution we see that the bandwidth of $f_2^2(t)$ is twice the bandwidth of $f_2(t)$, i.e., 300kHz. The Nyquist rate is therefore, 600kHz.

(iv) $f_1^3(t)$ **[2]**

Answer

The Fourier Transform of $f_1^3(t)$ is $\frac{1}{2\pi}\left(\frac{1}{2\pi}F_1(\omega) * F_1(\omega)\right) * F_1(\omega) = \frac{1}{4\pi^2}F_1(\omega) * F_1(\omega) * F_1(\omega)$. From the property regarding the size of convolution we see that the bandwidth of $f_1^3(t)$ is three times the bandwidth of $f_1(t)$, i.e., 300kHz. The Nyquist rate is therefore, 300kHz.

(v) $f_1(t)f_2^2(t)$ **[2]**

Answer

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Based on the above the bandwidth of $f_1(t)f_2^2(t)$ is the sum of the individual bandwidths, i.e., 400kHz. The Nyquist rate is therefore, 800kHz.

(g) Consider the discrete-time, causal LTI system with input $x[n]$ and output $y[n]$ related with the difference equation:

$$y[n] - \frac{9}{20}y[n-1] + \frac{1}{20}y[n-2] = x[n] - \frac{1}{4}x[n-1]$$

(i) Find the analytical expression and the Region of Convergence (ROC) of the $z-$transform of the impulse response of the above system. **[3]**

(ii) Find the analytical expression and the Region of Convergence (ROC) of the $z-$transform of the output if $x[n] = \left(\frac{1}{3}\right)^n u[n]$. **[2]**

Answer

(i)

$$Y(z) - \frac{9}{20}z^{-1}Y(z) + \frac{1}{20}z^{-2}Y(z) = X(z) - \frac{1}{4}z^{-1}X(z)$$

$$Y(z)(1 - \frac{9}{20}z^{-1}Y(z) + \frac{1}{20}z^{-2}) = X(z)(1 - \frac{1}{4}z^{-1})$$

$$Y(z)(1 - \frac{1}{4}z^{-1})(1 - \frac{1}{5}z^{-1}) = X(z)(1 - \frac{1}{4}z^{-1}) \Rightarrow$$

$$Y(z)\left(1 - \frac{1}{5}z^{-1}\right) = X(z) \Rightarrow \frac{Y(z)}{X(z)} = \frac{1}{\left(1 - \frac{1}{5}z^{-1}\right)} \Rightarrow H(z) = \frac{z}{z - 1/5}$$

$h[n] = \left(\frac{1}{5}\right)^n u[n]$, $|z| > 1/5$, causal system

(ii)

$X(z) = \frac{z}{z-1/3}$, $|z| > 1/3$

$Y(z) = \frac{z^2}{\left(z-\frac{1}{5}\right)\left(z-\frac{1}{3}\right)}$, ROC: $|z| > \frac{1}{5} \cap |z| > \frac{1}{3} = |z| > \frac{1}{3}$

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2. This question carries 30% of the mark.

(a) (i) Find the analytical expression and the Region of Convergence (ROC) of the Laplace transform of the continuous-time signal $x(t) = -ae^{-at}u(t) + e^{-at}\delta(t)$, with a real and positive and $u(t)$ the continuous-time unit step function. **[5]**

Answer

$$X(s) = -a\int_{-\infty}^{\infty} e^{-at}u(t)e^{-st}dt + \int_{-\infty}^{\infty} e^{-at}\delta(t)e^{-st}dt = -a\int_{0}^{\infty} e^{-at}e^{-st}dt + 1 =$$

$$\frac{e^{-(s+a)t}}{-(s+a)}\Big|_0^{\infty} + 1 = -a\frac{1}{s+a} + 1 = \frac{s}{s+a}, \text{ ROC: } \mathrm{Re}\{s\} > -a$$

(ii) Explain why the Laplace function is not adequate in order to determine the corresponding function in time. **[2]**

Answer

The Laplace transform function is not adequate since two functions in time can have the same Laplace transforms. The ROCs are required too.

(iii) Find the analytical expression and the Region of Convergence (ROC) of the Laplace transform of the signal $\delta(t-a)$, with a real and positive and $\delta(t)$ the continuous impulse function. **[3]**

Answer

$X(s) = \int_{-\infty}^{\infty} \delta(t-a)e^{-st}dt = e^{-sa}$, ROC: Entire plane

(iv) Find the analytical expression and the Region of Convergence (ROC) of Laplace transform of the signal $u(t)$, with $u(t)$ the continuous unit step function. **[2]**

Answer

$X(s) = \int_{-\infty}^{\infty} u(t)e^{-st}dt = \int_{0}^{\infty} e^{-st}dt = \frac{1}{s}$, ROC:

(b) Consider a continuous-time causal signal $x(t)$ with Laplace transform $X(s) = \frac{2s+5}{s^2+5s+6}$. Find the analytical expression of $x(t)$ and furthermore, find the region of convergence of $X(s)$. **[5]**

Answer

$$X(s) = \frac{1}{s+2} + \frac{1}{s+3} \Rightarrow$$
$$x(t) = e^{-2t}u(t) + e^{-3t}u(t)$$
ROC: $\mathrm{Re}\{s\} > -2$

(c) Consider a continuous, Linear Time-Invariant system with input the signal $x(t) = u(t)$, the unit step function and output the signal $y(t) = -\frac{2}{3}e^{t}u(-t) + \frac{1}{3}e^{-2t}u(t)$. Determine the transfer function of the system $H(s)$ and its Region of Convergence (ROC). **[5]**

Answer

$$y(t) = -\frac{2}{3}e^{t}u(-t) + \frac{1}{3}e^{-2t}u(t)$$
$$Y(s) = \frac{2}{3}\frac{1}{s-1} + \frac{1}{3}\frac{1}{s+2} = \frac{1}{3}\frac{3s+3}{(s-1)(s+2)} = \frac{s+1}{(s-1)(s+2)}$$
ROC: $\mathrm{Re}\{s\} < 1 \cap \mathrm{Re}\{s\} > -2 \Rightarrow -2 < \mathrm{Re}\{s\} < 1$
$$X(s) = \frac{1}{s}$$

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$$H(s) = \frac{Y(s)}{X(s)} = \frac{s(s+1)}{(s-1)(s+2)}$$

ROC: entire plane
Final ROC: $-2 < \text{Re}\{s\} < 1$

(d) The output $y(t)$ of an LTI system is related to the input $x(t)$ by the differential equation

$$\frac{d^2y(t)}{dt^2} + 11\frac{dy(t)}{dt} + 24y(t) = 5\frac{dx(t)}{dt} + 5x(t)$$

Let $X(s)$ and $Y(s)$ denote the Laplace transforms of $x(t)$ and $y(t)$, respectively, and let $H(s)$ denote the Laplace transform of the system's impulse response $h(t)$.

(i) Determine the transfer function $H(s)$ of the system. **[3]**

Answer

$(s^2 + 11s + 24)Y(s) = 5sX(s) + 5X(s) \Rightarrow (s+3)(s+8)Y(s) = (5s+5)X(s) \Longrightarrow$

$$\frac{Y(s)}{X(s)} = \frac{-2}{s+3} + \frac{7}{s+8} = H(s)$$

(ii) Determine all possible forms of the impulse response $h(t)$. **[3]**

Answer

$h(t) = -2e^{-3t}u(t) + 7e^{-8t}u(t)$, ROC: $\text{Re}\{s\} > -3$, stable and causal
$h(t) = -2e^{-3t}u(t) - 7e^{-8t}u(-t)$, ROC: $\text{Re}\{s\} > -3$, $\text{Re}\{s\} < -8$, not possible
$h(t) = 2e^{-3t}u(-t) + 7e^{-8t}u(t)$, ROC: $\text{Re}\{s\} > -8$, $\text{Re}\{s\} < -3$, neither stable nor causal
$h(t) = 2e^{-3t}u(-t) - 7e^{-8t}u(-t)$, ROC: $\text{Re}\{s\} < -8$, neither stable nor causal

(iii) Can the above system be both stable and causal? **[2]**

Answer

YES, first case above

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3. This question carries 30% of the mark.

(a) Show that the Fourier transform of the function $x(t) = \frac{W}{\pi}\text{sinc}(Wt)$, where W is a constant parameter is $X(\omega) = \text{rect}(\frac{\omega}{2W})$ with

$$\text{rect}(t) = \begin{cases} 1 & |t| < 0.5 \\ 0.5 & |t| = 0.5 \\ 0 & \text{otherwise} \end{cases}$$

Answer

$$x(t) = \frac{1}{2\pi}\int_{-\infty}^{\infty} X(\omega)e^{j\omega t}d\omega = \frac{1}{2\pi}\int_{-W}^{W} e^{j\omega t}d\omega = \frac{1}{2\pi}\left(\frac{e^{jWt}}{jt} - \frac{e^{-jWt}}{jt}\right) = \frac{2j\sin(Wt)}{2\pi jt}$$
$$= W\frac{\sin(Wt)}{\pi Wt} = \frac{W}{\pi}\text{sinc}(Wt)$$

(b) Determine the Nyquist sampling rate and the Nyquist sampling interval for the following signals.

(i) $x_1(t) = \text{sinc}^2(100\pi t)$. **[3]**

Answer

From the Fourier transform of $\text{sinc}(100\pi t)$ we see that the bandwidth of it is 50 Hz. Therefore, the bandwidth of $\text{sinc}^2(100\pi t)$ is 100Hz.

(ii) $x_2(t) = 0.01\text{sinc}^2(100\pi t)$. **[3]**

Answer

Same as in (i)

(iii) $x_3(t) = \text{sinc}(100\pi t) + 3\text{sinc}^2(60\pi t)$. **[3]**

Answer

The bandwidth of $\text{sinc}(100\pi t)$ is 50Hz and the bandwidth of $\text{sinc}^2(60\pi t)$ is $30 \times 2 = 60$Hz and therefore, the total bandwidth is 60Hz.

(iv) $x_4(t) = \text{sinc}(50\pi t)\text{sinc}(100\pi t)$. **[3]**

Answer

The bandwidth of $\text{sinc}(50\pi t)$ is 25Hz and the bandwidth of $\text{sinc}(100\pi t)$ is 50Hz and therefore, the total bandwidth is 75Hz.

(c) (i) Consider a discrete signal $x[n]$ with $z-$transform $X(z)$. Find the $z-$transforms of the signals $x_1[n] = x[n-2]$ and $x_2[n] = x[-n-3]$ as functions of the $z-$ transform $X(z)$. **[3]**

Answer

Assume that $x_1[n] = x[n-2]$.

$$X_1(z) = \sum_{n=-\infty}^{\infty} x_1[n]z^{-n} = \sum_{n=-\infty}^{\infty} x[n-2]z^{-n} = \sum_{n=-\infty}^{\infty} x[n-2]z^{-(n-2)}\,z^{-2}$$
$$= z^{-2}X(z)$$

Assume that $x_2[n] = x[-n-3]$.

$$X_2(z) = \sum_{n=-\infty}^{\infty} x_2[n]z^{-n} = \sum_{n=-\infty}^{\infty} x[-n-3]z^{-n} = \sum_{n=-\infty}^{\infty} x[-n-3]\left(\frac{1}{z}\right)^{n}\left(\frac{1}{z}\right)^{-3}\left(\frac{1}{z}\right)^{3}$$
$$= \left(\frac{1}{z}\right)^{-3}\sum_{n=-\infty}^{\infty} x[-n-3]\left(\frac{1}{z}\right)^{-(-n-3)} = \left(\frac{1}{z}\right)^{-3}\sum_{k=\infty}^{-\infty} x[k]\left(\frac{1}{z}\right)^{-k} = \left(\frac{1}{z}\right)^{-3}X\left(\frac{1}{z}\right)$$

(ii) Consider a signal $y[n]$ which is related to two signals $x_1[n]$ and $x_2[n]$ by

$$y[n] = x_1[n] * x_2[n]$$

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where

$$x_1[n] = \left(\frac{1}{2}\right)^n u[n] \text{ and } x_2[n] = \left(\frac{1}{3}\right)^n u[n]$$

with $u[n]$ the discrete unit step function. Determine the $z-$ transform $Y(z)$ of $y[n]$.with $u[n]$ the discrete unit step function. Use the results obtained in question (b)(i) above to determine the $z-$ transform of $y[n]$. **[4]**

Answer

$$Y(z) = Z\{x_1[n]\} \cdot Z\{x_2[n]\} = z^2 \frac{1}{\left(z-\frac{1}{2}\right)} \cdot \frac{1}{\left(z-\frac{1}{3}\right)}$$

(d) Consider an odd sequence $x[n]$ (i.e., $x[n] = -x[-n]$) with rational $z-$ transform $X(z)$.

(i) Find $X\left(\frac{1}{z}\right)$ as a function of $X(z)$. **[3]**

Answer

$g[n] = -x[-n]$

$G(z) = \sum_{n=-\infty}^{\infty} g[n]z^{-n} = \sum_{n=-\infty}^{\infty} -x[-n]z^{-n} =$

$\sum_{n=-\infty}^{\infty} -x[n]\, z^{n} = \sum_{n=-\infty}^{\infty} -x[n]\left(\frac{1}{z}\right)^{-n}$

$= -X\left(\frac{1}{z}\right)$

Therefore, it is obvious that if $x[n] = -x[-n]$ then $X(z) = -X\left(\frac{1}{z}\right)$.

(ii) From your result in part (d)(i), show that if a pole (zero) of $X(z)$ occurs at $z = z_0$, then a pole (zero) must also occur at $z = 1/z_0$. **[3]**

Answer

Assume that $x[n] = -x[-n]$ which implies $X(z) = -X\left(\frac{1}{z}\right)$.

Assume that

$X(z) = P(z)(z - z_0)$

$-X\left(\frac{1}{z}\right) = -P\left(\frac{1}{z}\right)\left(\frac{1}{z} - z_0\right) = -P\left(\frac{1}{z}\right)(1 - zz_0)\frac{1}{z}$

For $z = \frac{1}{z_0}$, $X\left(\frac{1}{z}\right) = 0 \Rightarrow \frac{1}{z_0}$ is a zero for $X\left(\frac{1}{z}\right)$.

Same analysis is valid for poles.

(iii) Verify the results in part (d)(i) the sequence $\delta[n+2] - \delta[n-2]$ **[3]**

Answer

$Z\{\delta[n+2]\} = z^2$

$Z\{\delta[n-2]\} = z^{-2}$

$X(z) = Z\{\delta[n+2]\} - Z\{\delta[n-2]\} = z^2 - z^{-2}$

$X\left(\frac{1}{z}\right) = -X(z)$

$$X(z) = \frac{z^4 - 1}{z^2} = \frac{(z^2-1)(z^2+1)}{z^2} = \frac{(z-1)(z+1)(z-i)(z+i)}{z^2}$$

We see that we have poles at 0 and ∞.

We see that we have zeros at 1 and -1. (1 is the same as $1/1$ and also -1 is the same as $-1/1$).

Another zero is i. We see that $\frac{1}{i} = -i$ which is also a zero. The same is valid for $-i$.

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